Heart Disease Modeling in R

Technical Summary Report

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1. Introduction

The goal of this project is to create several models that will efficiently predict whether an individual is at risk of heart disease and their maximum heart rate. To accomplish this, this project examines relationships between patient health indicators and the recorded presence of heart disease using logistic regression and random forest models in R. The dataset contains 303 observations and includes demographic information, clinical measurements, symptoms, and diagnostic indicators.

Two logistic regression models are used to estimate the probability of heart disease. Their fit and performance are examined using coefficient tests, the Hosmer–Lemeshow goodness-of-fit test, confusion matrices, and receiver operating characteristic curves. A classification random forest is also used to predict heart disease status, while a regression random forest is used to predict maximum heart rate achieved. The logistic regression results are evaluated on the same observations used to fit the models, while the random forest results include separate training and validation sets.

With further development, these models have the capacity to analyze medical records and detect subtle risks that might otherwise go unnoticed by human doctors, enhancing the accuracy and comprehensiveness of medical evaluations.

2. Data Preparation

age	sex	cp	trestbps	chol	fbs	restecg	thalach	exang	oldpeak	slope	ca	thal	target
<int>	<fct>	<fct>	<int>	<int>	<fct>	<fct>	<int>	<fct>	<dbl>	<fct>	<fct>	<fct>	<fct>
62	1	2	130	231	0	1	146	0	1.8	1	3	3	1
58	0	0	130	197	0	1	131	0	0.6	1	0	2	1
60	0	3	150	240	0	1	171	0	0.9	2	0	2	1
63	1	0	140	187	0	0	144	1	4.0	2	2	3	0
62	1	0	120	267	0	1	99	1	1.8	1	2	3	0
63	0	2	135	252	0	0	172	0	0.0	2	0	2	1
43	1	0	150	247	0	1	171	0	1.5	2	0	2	1
42	1	2	120	240	1	1	194	0	0.8	0	0	3	1
59	1	2	126	218	1	1	134	0	2.2	1	1	1	0
48	1	0	124	274	0	0	166	0	0.5	1	0	3	0

The data set has 303 rows and 14 columns. Above are the first 10 rows of the data.
The variables I will be working with today are:

- age – the person's age in years
- sex – the person's sex (1 = male, 0 = female)
- cp – the type of chest pain experienced (0=no pain, 1=typical angina, 2=atypical angina, 3=non-anginal pain)
- trestbps – the person's resting blood pressure
- chol – the person's cholesterol measurement in mg/dl
- restecg – resting electrocardiographic measurement (0=normal, 1=having ST-T wave abnormality, 2=showing probable or definite left ventricular hypertrophy by Estes' criteria)
- thalach – the person's maximum heart rate achieved
- exang – exercise-induced angina (1=yes, 0=no)
- ca – the number of major vessels (0-3)
- target – heart disease (0=no, 1=yes)

3. Model #1 - First Logistic Regression Model

Reporting Results

Logistic regression model.

The general form of the logistic multiple regression model for heart disease (target) using variables age (age), resting blood pressure (trestbps), exercise-induced angina (exang), and maximum heart rate achieved (thalach) (Berrier et al., 2016):

$$E(y) = \pi(x) = \frac{e^{(\beta_0+\beta_1x_1+\beta_2x_2+\beta_3x_3+\beta_4x_4)}}{1+e^{(\beta_0+\beta_1x_1+\beta_2x_2+\beta_3x_3+\beta_4x_4)}},$$

where **y** is 1 for heart disease and 0 for no heart disease, $\mathbf{x_1}$ – age, $\mathbf{x_2}$ – resting blood pressure, $\mathbf{x_3}$ – presence of exercise-induced angina and $\mathbf{x_4}$ – maximum heart rate achieved.

This model can also be expressed as a log-odds function:

$$\ln\left(\frac{\pi}{1-\pi}\right) = \beta_0 + \beta_1x_1 + \beta_2x_2 + \beta_3x_3 + \beta_4x_4$$

$\boldsymbol{\pi}$ – the probability that the individual will have a heart disease based on the above-mentioned variables.

$\frac{\boldsymbol{\pi}}{\mathbf{1-\pi}}$ – the odds of an individual having a heart disease.

The prediction regression model in terms of the natural log of odds is:

$$\ln(odds) = \widehat{\beta_0} + \widehat{\beta_1}x_1 + \widehat{\beta_2}x_2 + \widehat{\beta_3}x_3 + \widehat{\beta_4}x_4,$$

where ***odds*** – the odds of heart disease (target=1), $\mathbf{x_1}$ – age, $\mathbf{x_2}$ – resting blood pressure, $\mathbf{x_3}$ – presence of exercise-induced and $\mathbf{x_4}$ – maximum heart rate achieved.

The output from R regarding the logistic regression model:

```
Call:
glm(formula = target ~ age + trestbps + exang + thalach, family = "binomial",
    data = heart_data)

Deviance Residuals:
    Min       1Q   Median       3Q      Max
-2.0935  -0.7944   0.4954   0.8133   2.2343

Coefficients:
             Estimate Std. Error z value Pr(>|z|)
(Intercept) -1.021121   1.784194  -0.572   0.5671
age         -0.017549   0.017144  -1.024   0.3060
trestbps    -0.014888   0.008337  -1.786   0.0741 .
exang1      -1.624981   0.305774  -5.314 1.07e-07 ***
thalach      0.031095   0.007275   4.274 1.92e-05 ***
```

From this output, the prediction model equation in terms of natural log of odds is:

$$\ln(odds) = -1.0211 - 0.0175x_1 - 0.0149x_2 - 1.6250x_3 + 0.0311x_4,$$

where ***odds*** – the odds of heart disease (target=1), $\mathbf{x_1}$ – age, $\mathbf{x_2}$ – resting blood pressure, $\mathbf{x_3}$ – presence of exercise-induced angina and $\mathbf{x_4}$ – maximum heart rate achieved.

Estimated coefficient of maximum heart rate achieved

The estimated coefficient for maximum heart rate achieved is 0.0311. Holding age, resting blood pressure, and exercise-induced angina constant, a one-beat-per-minute increase in maximum heart rate is associated with an increase of 0.0311 in the estimated log-odds of heart disease. Expressed as an odds ratio, $e^{0.0311} \approx \mathbf{1.032}$, corresponding to approximately **3.2%** higher estimated odds for each additional beat per minute ($(1.0316 - 1) \times 100 = 3.16\%$).

Evaluating Model Significance

Hosmer-Lemeshow goodness of fit (GOF) test

The following image is the output form R regarding the Hosmer-Lemeshow goodness of fit test:

```
        Hosmer and Lemeshow goodness of fit (GOF) test

data:  logit1$y, fitted(logit1)
X-squared = 44.622, df = 48, p-value = 0.612
```

Null hypothesis, H_0: the model fits the data.
Alternative Hypothesis, H_a: the model does not fit the data.
The **Chi-squared** statistic is **44.622** with a corresponding **p**-value of **0.612**. Since the **p**-value is larger than the significance level $\boldsymbol{\alpha = 0.05}$, we fail to reject the null hypothesis, which suggests that the test provides no evidence of lack of fit.

Wald's test (z-value)

```
Coefficients:
             Estimate Std. Error z value Pr(>|z|)
(Intercept) -1.021121   1.784194  -0.572   0.5671
age         -0.017549   0.017144  -1.024   0.3060
trestbps    -0.014888   0.008337  -1.786   0.0741 .
exang1      -1.624981   0.305774  -5.314 1.07e-07 ***
thalach      0.031095   0.007275   4.274 1.92e-05 ***
---
Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
```

For each predictor, the Wald test evaluates $\boldsymbol{H_o}$: $\boldsymbol{\beta = 0}$ against $\boldsymbol{H_a}$: $\boldsymbol{\beta \neq 0}$. A term is considered statistically significant when its p-value is less than **0.05**.

PREDICTOR	ESTIMATE	Z-VALUE	P-VALUE	RESULT
AGE	−0.0175	−1.024	0.3060	Not significant
RESTING BLOOD PRESSURE	−0.0149	−1.786	0.0741	Not significant
EXERCISE-INDUCED ANGINA: YES VS. NO	−1.6250	−5.314	<0.0001	Significant
MAXIMUM HEART RATE ACHIEVED	0.0311	4.274	<0.0001	Significant

At the 5% significance level, exercise-induced angina and maximum heart rate achieved are statistically significant, while age and resting blood pressure are not. A nonsignificant result does not establish that a variable is medically unimportant; it means that this model and sample do not provide sufficient evidence of an association after accounting for the other predictors.

The negative coefficient for exercise-induced angina is counterintuitive clinically. However, the coefficient represents a conditional association within this dataset and may be influenced by variable coding, correlations among predictors, sampling variation, or variables not included in the model.

Confusion matrix

	Prediction = 0	Prediction = 1
Actual = 0	True Negatives	False Positives
Actual = 1	False Negatives	True Positives

	Prediction: target=0	Prediction: target=1
Actual: target=0	89	49
Actual: target=1	31	134

From this R output we can see that:

- true negatives = 89,
- false negatives = 31,
- false positives = 49,
- true positives = 134.

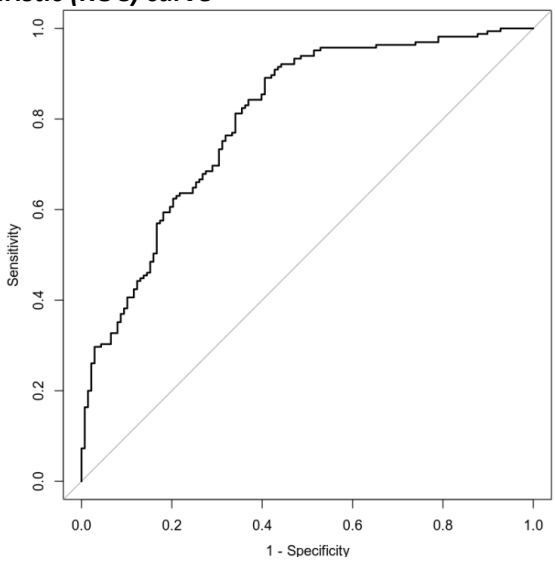
$$\textbf{Accuracy} = \frac{TP+TN}{TP+TN+FP+FN} = \frac{134+89}{134+89+49+31} = \frac{223}{303} = 0.7360$$

$$\textbf{Precision} = \frac{TP}{TP+FP} = \frac{134}{134+49} = \frac{134}{183} = 0.7322$$

$$\textbf{Recall} \text{ (Sensitivity/True Positive Rate)} = \frac{TP}{TP+FN} = \frac{134}{134+31} = \frac{134}{165} = 0.8121$$

The model correctly classifies **223 of the 303** observations, giving an accuracy of **73.60%.** Among the observations predicted as heart disease**, 73.22%** are true positives, which is the model's precision. The model identifies **81.21%** of the observations recorded as heart disease, which is its recall or sensitivity. These results describe performance on the same dataset used to fit the model, so they may be more optimistic than performance on new patients.

Receiver Operating Characteristic (ROC) curve



The ROC curve shows the model's ability to distinguish between observations with and without recorded heart disease over different classification thresholds. A curve closer to the upper-left corner represents better discrimination, while the diagonal line represents performance no better than random classification. The curve remains above the diagonal for most thresholds, indicating useful—but not perfect—discrimination.

```
[1] "Area Under the Curve (AUC)"

0.8007
```

AUC (area under the curve) **= 0.8007**. This means that, when comparing one randomly selected observation from each outcome group, the model would assign a higher predicted probability to the heart-disease observation approximately **80%** of the time.

Making Predictions Using Model

```
[1] "Prediction: 50 years old, resting blood pressure of 122, exercise induced angina, maximum heart r
ate of 140"

1: 0.2716
```

The probability of an individual having heart disease who is 50 years old, has a resting blood pressure of 122, has exercise induced angina, and has maximum heart rate of 140 is **0.2716**.
The odds of this event occurring is:

$$\text{Odds} = \frac{\pi}{1-\pi} = \frac{0.2716}{1-0.2716} = \mathbf{0.37}$$

This means that the odds of an individual having heart disease who is 50 years old, has a resting blood pressure of 122, has exercise induced angina, and has maximum heart rate of 140 is about **0.37 to 1**.

```
[1] "Prediction: 50 years old, resting blood pressure of 130, no exercise induced angina, maximum hear
t rate of 165"

1: 0.7853
```

The probability of an individual having heart disease who is 50 years old, has a resting blood pressure of 130, does not have an exercise induced angina, and has maximum heart rate of 165 is **0.7853**.
The odds of this event occurring is:

$$\text{Odds} = \frac{\pi}{1-\pi} = \frac{0.7853}{1-0.7853} = \mathbf{3.66}$$

This means that the odds of an individual having heart disease who is 50 years old, has a resting blood pressure of 130, does not have an exercise induced angina, and has maximum heart rate of 165 is about **3.66 to 1**.

Considering that both individuals are of the same age, we can see how other variables affect the presence of heart disease. The first individual is more likely to not have heart disease, probably due to lower resting blood pressure and maximum heart achieved. The second individual is more likely to have heart disease according to the prediction. However, considering that this individual does not have an exercise-induced angina, the odds should be lower.

4. Model #2 - Second Logistic Regression Model

Reporting Results

Logistic regression model.

The general form of the logistic multiple regression model for heart disease (target) using variables age (age), resting blood pressure (trestbps), type of chest pain experienced (cp), maximum heart rate achieved (thalach), including the quadratic term for age and the interaction term between age and maximum heart rate achieved:

$$E(y) = \pi(x) = \frac{e^{(\beta_0+\beta_1x_1+\beta_2x_2+\beta_3x_3+\beta_4x_4+\beta_5x_5+\beta_6x_6+\beta_7x_1^2+\beta_8x_1x_6)}}{1+e^{(\beta_0+\beta_1x_1+\beta_2x_2+\beta_3x_3+\beta_4x_4+\beta_5x_5+\beta_6x_6+\beta_7x_1^2+\beta_8x_1x_6)}},$$

where **y** is 1 for heart disease and 0 for no heart disease, $\mathbf{x_1}$ – age, $\mathbf{x_2}$ – resting blood pressure, $\mathbf{x_3}$ – typical angina, $\mathbf{x_4}$ – atypical angina, $\mathbf{x_5}$ – non-anginal pain and $\mathbf{x_6}$ – maximum heart rate achieved.

The prediction regression model in terms of the natural log of odds is:

$$\ln(odds) = \widehat{\beta_0} + \widehat{\beta_1}x_1 + \widehat{\beta_2}x_2 + \widehat{\beta_3}x_3 + \widehat{\beta_4}x_4 + \widehat{\beta_5}x_5 + \widehat{\beta_6}x_6 + \widehat{\beta_7}x_1^2 + \widehat{\beta_8}x_1x_6,$$

where ***odds*** – the odds of heart disease (target=1), $\mathbf{x_1}$ – age, $\mathbf{x_2}$ – resting blood pressure, $\mathbf{x_3}$ – typical angina, $\mathbf{x_4}$ – atypical angina, $\mathbf{x_5}$ – non-anginal pain and $\mathbf{x_6}$ – maximum heart rate achieved.

The output from R regarding the logistic regression model:

```
Call:
glm(formula = target ~ age + trestbps + cp + thalach + I(age^2) +
    age:thalach, family = "binomial", data = heart_data)

Deviance Residuals:
    Min       1Q   Median       3Q      Max
-2.6961  -0.7537   0.2925   0.7123   2.3058

Coefficients:
              Estimate Std. Error z value Pr(>|z|)
(Intercept) -1.556e+01  1.054e+01  -1.476  0.13988
age          1.744e-01  2.669e-01   0.653  0.51357
trestbps    -1.958e-02  8.978e-03  -2.181  0.02916 *
cp1          1.913e+00  4.437e-01   4.313 1.61e-05 ***
cp2          2.037e+00  3.473e-01   5.867 4.45e-09 ***
cp3          1.777e+00  5.477e-01   3.245  0.00117 **
thalach      1.363e-01  5.119e-02   2.663  0.00775 **
I(age^2)     8.424e-04  1.750e-03   0.481  0.63025
age:thalach -1.867e-03  8.909e-04  -2.095  0.03616 *
```

From this output, the prediction model equation in terms of natural log of odds is:

$$\ln(odds) = -15.56 - 0.1744x_1 - 0.0196x_2 + 1.913x_3 + 2.037x_4 + 1.777x_5 + 0.1363x_6 + 0.0008x_1^2 - 0.0019x_1x_6,$$

where ***odds*** – the odds of heart disease (target=1), $\mathbf{x_1}$ – age, $\mathbf{x_2}$ – resting blood pressure, $\mathbf{x_3}$ – typical angina, $\mathbf{x_4}$ – atypical angina, $\mathbf{x_5}$ – non-anginal pain and $\mathbf{x_6}$ – maximum heart rate achieved.

Evaluating Model Significance

Hosmer-Lemeshow goodness of fit (GOF) test

The following image is the output form R regarding the Hosmer-Lemeshow goodness of fit test:

```
        Hosmer and Lemeshow goodness of fit (GOF) test

data:  logit2$y, fitted(logit2)
X-squared = 52, df = 48, p-value = 0.3209
```

Null hypothesis, H_0: the model fits the data.

Alternative Hypothesis, H_a: the model does not fit the data.

The **Chi-squared** statistic is **52** with a corresponding **p**-value of **0.3209**. Since the **p**-value is larger than the significance level $\alpha = 0.05$, we fail to reject the null hypothesis, which suggests that the test provides no evidence of lack of fit.

Wald's test

```
Coefficients:
              Estimate Std. Error z value Pr(>|z|)
(Intercept) -1.556e+01  1.054e+01  -1.476  0.13988
age          1.744e-01  2.669e-01   0.653  0.51357
trestbps    -1.958e-02  8.978e-03  -2.181  0.02916 *
cp1          1.913e+00  4.437e-01   4.313 1.61e-05 ***
cp2          2.037e+00  3.473e-01   5.867 4.45e-09 ***
cp3          1.777e+00  5.477e-01   3.245  0.00117 **
thalach      1.363e-01  5.119e-02   2.663  0.00775 **
I(age^2)     8.424e-04  1.750e-03   0.481  0.63025
age:thalach -1.867e-03  8.909e-04  -2.095  0.03616 *
---
Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
```

For each predictor, the Wald test evaluates H_o: $\beta = 0$ against H_a: $\beta \neq 0$. A term is considered statistically significant when its p-value is less than **0.05**.

PREDICTOR	ESTIMATE	Z-VALUE	P-VALUE	RESULT
AGE	0.1744	0.653	0.5136	Not significant
RESTING BLOOD PRESSURE	−0.0196	−2.181	0.0292	Significant
TYPICAL ANGINA VS. NO CHEST PAIN	1.9130	4.313	<0.0001	Significant
ATYPICAL ANGINA VS. NO CHEST PAIN	2.0370	5.867	<0.0001	Significant
NON-ANGINAL PAIN VS. NO CHEST PAIN	1.7770	3.245	0.0012	Significant
MAXIMUM HEART RATE ACHIEVED	0.1363	2.663	0.0078	Significant
AGE SQUARED	0.0008	0.481	0.6303	Not significant
AGE × MAXIMUM HEART RATE	−0.0019	−2.095	0.0362	Significant

At the 5% significance level, resting blood pressure, all three chest-pain categories, maximum heart rate, and the interaction between age and maximum heart rate are statistically significant. Age and age squared are not individually significant.

Because this model contains an interaction and a quadratic term, the age and maximum-heart-rate coefficients should not be interpreted independently. The association between maximum heart rate and the outcome changes with age, while the modeled association involving age also depends on maximum heart rate and the quadratic age term.

Confusion matrix

	Prediction: target=0	Prediction: target=1
Actual: target=0	102	36
Actual: target=1	36	129

From this R output we can see that:

- true negatives = 102,
- false negatives = 36,
- false positives = 36,
- true positives = 129.

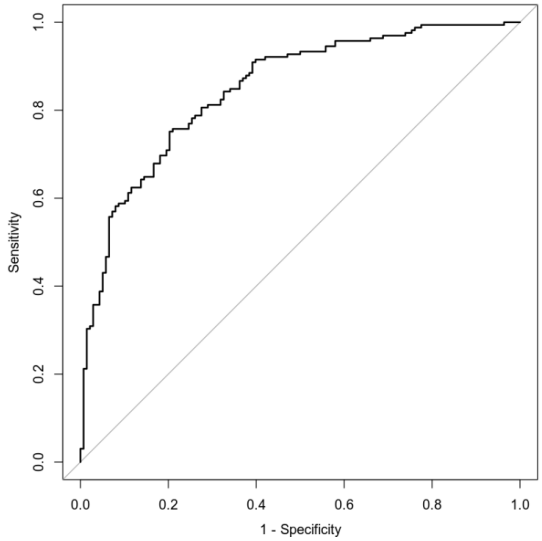
Accuracy = $\frac{TP+TN}{TP+TN+FP+FN} = \frac{102+129}{102+129+36+36} = \frac{231}{303} = 0.7624$

Precision = $\frac{TP}{TP+FP} = \frac{129}{129+36} = \frac{129}{165} = 0.7818$

Recall (Sensitivity/True Positive Rate) = $\frac{TP}{TP+FN} = \frac{129}{129+36} = \frac{129}{165} = 0.7818$

The model correctly classifies 231 of the 303 observations, resulting in an accuracy of 76.24%. Precision is 78.18%, meaning that approximately 78% of the observations predicted as heart disease are true positives. Recall is also 78.18%, meaning that the model identifies approximately 78% of the observations recorded as heart disease.

Receiver Operating Characteristic (ROC) curve



The ROC curve remains above the diagonal for most thresholds and bends toward the upper-left corner, indicating that the model distinguishes between the two outcome groups better than random classification.

```
[1] "Area Under the Curve (AUC)"

0.8478
```

AUC (area under the curve) **= 0.8478**. This value indicates that the model is good at predicting heart disease and no heart disease.

Making Predictions Using Model

```
[1] "Prediction: 50 years old, resting blood pressure of 115, no chest pain, maximum heart rate of 13
3"

1: 0.2188
```

The probability of an individual having heart disease who is 50 years old, has a resting blood pressure of 115, has no chest pain, and has maximum heart rate of 133 is **0.2188**.
The odds of this event occurring is:

$$\text{Odds} = \frac{\pi}{1-\pi} = \frac{0.2188}{1-0.2188} = 0.28$$

This means that the odds of an individual having heart disease who is 50 years old, has a resting blood pressure of 115, has no chest pain, and has maximum heart rate of 133 is about **0.28 to 1**. This individual is more likely to not have heart disease.

```
[1] "Prediction: 50 years old, resting blood pressure of 125, typical angina, maximum heart rate of 15
5"

1: 0.8007
```

The probability of an individual having heart disease who is 50 years old, has a resting blood pressure of 125, has typical angina, and has maximum heart rate of 155 is **0.8007**.
The odds of this event occurring is:

$$\text{Odds} = \frac{\pi}{1-\pi} = \frac{0.8007}{1-0.8007} = 4.02$$

This means that the odds of an individual having heart disease who is 50 years old, has a resting blood pressure of 125, has typical angina, and has maximum heart rate of 155 is about **4.02 to 1**. This individual is more likely to have heart disease.

Considering that both individuals are of the same age, we can see how other variables affect the presence of heart disease. The first individual is more likely not to have heart disease, probably due to lower resting blood pressure and maximum heart achieved and no chest pain. The second individual is more likely to have heart disease. The odds of this individual having heart disease being 4 to 1 is quite reasonable, as they experience higher resting blood pressure, higher maximum heart rate achieved and typical angina.

5. Random Forest Classification Model

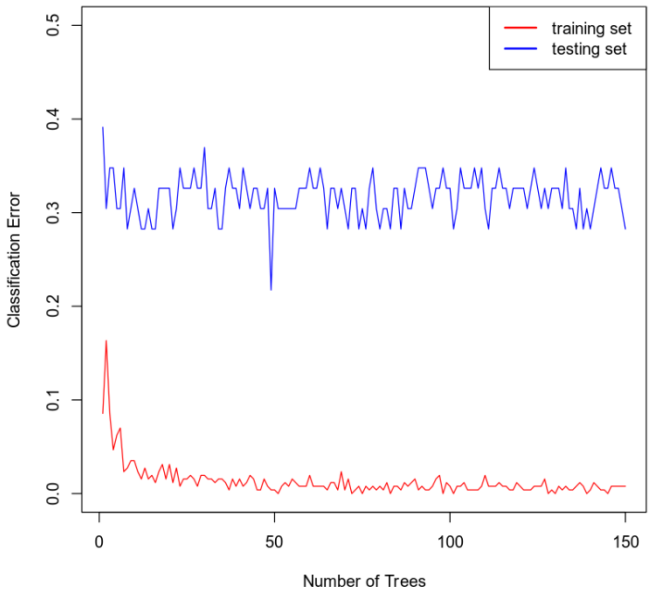
Reporting Results

To ensure the model performs well, first, I will split the heart_data data set into training and validation sets using 85% and 15% split, respectively. Since the data is small, 85%-15% is a good split.

```
[1] "Number of rows for the training set"
257
[1] "Number of rows for the testing set"
46
```

The original data set has 303 rows. The training set, which is the set of data from which the model learns the patterns has 257 rows, and the validation set, on which the trained model is tested has 46 rows.

Next, I will graph the training and testing error against the number of trees using a classification random forest model for the presence of heart disease (target) using variables age (age), sex (sex), chest pain type (cp), resting blood pressure (trestbps), cholesterol measurement (chol), resting electrocardiographic measurement (restecg), exercise-induced angina (exang), and number of major vessels (ca) using a maximum of 150 trees.



To find the optimal number of trees to use in our random forest model, we look at the training set line. As this line approaches zero, it means that there are very little to no classification errors. The optimal number of trees then would be the number after which the errors are zero or close to zero. As we can see from the graph above, the optimal number of trees is about 20 trees.

Evaluating the Utility of the model

Using the optimal number of trees, which we identified to be 20 trees, we will create a classification random forest model for the presence of heart disease using the same variables mentioned in the previous section. Based on this model, we will obtain two confusion matrices, for the training and for the testing set and evaluate their accuracy, precision, and recall.

Training set confusion matrix

```
[1] "Confusion Matrix: TRAINING set based on random forest model built using 20 trees"
```

A matrix: 2 × 2 of type chr

	Prediction: 0	Prediction: 1
Actual: 0	118	2
Actual: 1	0	137

From this R output we can see that:

- true negatives = 118,
- false negatives = 0,
- false positives = 2,
- true positives = 137.

$$\textbf{Accuracy} = \frac{TP+TN}{TP+TN+FP+FN} = \frac{137+118}{137+118+2+0} = \frac{255}{257} = 0.9922$$

$$\textbf{Precision} = \frac{TP}{TP+FP} = \frac{129}{129+36} = \frac{137}{137+2} = 0.9856$$

$$\textbf{Recall} \text{ (Sensitivity/True Positive Rate)} = \frac{TP}{TP+FN} = \frac{137}{137+0} = \frac{137}{137} = 1$$

The model correctly classifies 255 of the 257 training observations, producing 99.22% accuracy, 98.56% precision, and 100% recall. Because all these metrics are substantially high, we can confidently conclude that the model classifies the data it was trained on very well. However, they do not, by themselves, demonstrate that the model will perform equally well on unseen data.

Validation set confusion matrix

```
[1] "Confusion Matrix: TESTING set based on random forest model built using 20 trees"
```

A matrix: 2 × 2 of type chr

	Prediction: 0	Prediction: 1
Actual: 0	11	7
Actual: 1	8	20

From this R output we can see that:

- true negatives =11,
- false negatives = 8,
- false positives = 7,
- true positives = 20.

Accuracy = $\frac{TP+TN}{TP+TN+FP+FN} = \frac{20+11}{20+11+7+8} = \frac{31}{46} = 0.6739$

Precision = $\frac{TP}{TP+FP} = \frac{20}{20+7} = \frac{20}{27} = 0.7407$

Recall (Sensitivity/True Positive Rate) = $\frac{TP}{TP+FN} = \frac{20}{20+8} = \frac{20}{28} = 0.7143$

On the 46 validation observations, accuracy decreases to 67.39%, precision to 74.07%, and recall to 71.43%. The substantial difference between training and validation performance suggests that the model does not generalize as well as its training results imply. The small validation sample also makes these percentages relatively unstable.

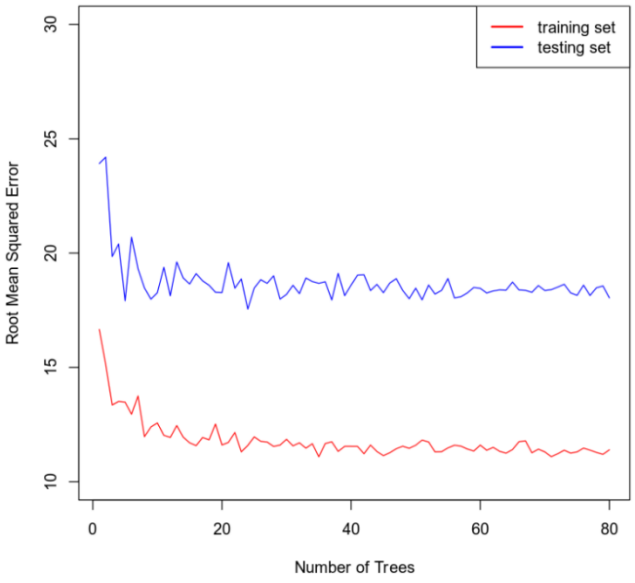
6. Random Forest Regression Model

Reporting Results

To ensure the model performs well, first, I will split the heart_data data set into training and validation sets using 80% and 20% split, respectively. Since the data is small, 80%-20% is a good split.

```
[1] "Number of rows for the training set"
242
[1] "Number of rows for the testing set"
61
```

The original data set has 303 rows. The training set, which is the set of data from which the model learns the patterns has 242 rows, and the validation set, on which the trained model is tested has 61 rows.



Next, I will graph the mean squared error against the number of trees for a random forest regression model for maximum heart rate achieved(thalach) using age (age), sex (sex), chest pain type (cp), resting blood pressure (trestbps), cholesterol measurement (chol), resting electrocardiographic measurement (restecg), exercise-induced angina (exang), and number of major vessels (ca) using a maximum of 80 trees.

To find the optimal number of trees to use in our random forest model, we look at the training set line. The training RMSE decreases rapidly as trees are added and then changes more gradually after approximately 20 trees.

Evaluating the Utility of the Random Forest Regression Model

Using the optimal number of trees, which we identified to be 20 trees, we will create a random forest regression model for maximum heart rate achieved using the same variables mentioned in the previous section. Based on this model, we will obtain the root mean squared errors for the training and for the testing set.

Training set RMSE

```
[1] "Root Mean Squared Error: TRAINING set based on random forest model built using 20 trees"

11.7762979145166
```

The Root Mean Squared Error (RMSE) is the standard deviation of prediction of residuals (Berrier et al., 2016). As seen from the R output above, the RMSE for the training set is **11.7763,** meaning that predictions for the training observations differ from the recorded values by approximately **11.78** beats per minute on a typical RMSE scale.

Validation set RMSE

```
[1] "Root Mean Squared Error: TESTING set based on random forest model built using 20 trees"

18.7266873327815
```

The validation RMSE is **18.73** beats per minute, which is notably higher than the training RMSE. This gap suggests weaker performance on unseen observations and possible overfitting.

7. Conclusion

Of the two logistic regression models, Model 2 performed better on the existing dataset. Its accuracy was 76.24%, compared with 73.60% for Model 1, and its AUC was 0.8478, compared with 0.8007 for Model 1. However, because both logistic models were evaluated using the same data on which they were fitted, these results do not establish how well Model 2 would perform on new patients.

The classification random forest fits the training data extremely well but performs substantially worse on the validation set, where its accuracy is 67.39%. This gap suggests limited generalization and possible overfitting. Its validation accuracy should not be compared directly with the logistic models' in-sample accuracies because the models were evaluated under different conditions. A fair comparison would require evaluating all classification models using the same training and validation partitions or the same cross-validation procedure.

The regression random forest produces a training RMSE of 11.78 beats per minute and a validation RMSE of 18.73 beats per minute. The higher validation error indicates that predictions are less accurate for unseen observations. Comparison with a baseline prediction method would be needed to determine whether this error represents a meaningful improvement.

These models, with further development, could analyze medical records to detect subtle, previously unnoticed risks, leading to more accurate and comprehensive medical evaluations. By identifying hidden patterns and correlations, they could enhance early disease detection and preventive care strategies. Additionally, they could personalize treatment plans by considering a patient's unique medical history and risk factors, ultimately improving patient outcomes and quality of care.

8. Citations

Berrier, H., Chan, C., & Pardoe, I. (2016). Applied Statistics with Data Analytics (R). zyBooks.com.